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# Reproducing “Bilinear MLPs Enable Weight-Based Mechanistic Interpretability”

## Abstract

Mechanistic interpretability typically relies on post-hoc analysis of model activations, but bilinear MLPs offer an alternative: architectures whose weights are directly interpretable through eigendecomposition of interaction tensors. We reproduce both main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST/Fashion-MNIST, and Section 5 (Language) discovering sentiment negation circuits via Sparse Autoencoder analysis. In vision, we fully reproduce the core findings: weight decay reduces effective rank from 38.5 to 15.5, confirming that regularization induces interpretable low-rank structure, while eigenvectors under noise augmentation show recognizable digit patterns. Our ablation disentangles noise and weight decay, finding that weight decay alone achieves the lowest rank while noise primarily improves accuracy. In language, we confirm negation circuit discovery—SAE features encoding “not + positive” and “not + negative” emerge as semantic opposites (cosine similarity  $-0.16$ ). However, we do *not* fully reproduce the low-rank interaction claim: the fraction of features achieving  $>0.75$  correlation varies from 32% (ts-medium) to 65% (fw-small). We interpret “most features” as a median threshold ( $>50\%$ ): fw-small (65%) reproduces the claim, while ts-medium (32%) and fw-medium (45%) fall short. We trace this discrepancy to both SAE training time and model training compute (tokens/parameter): comparing five SAE versions, correlation improves  $2.5\times$  with longer training, while models with higher tokens/param show better low-rank behavior. Additionally, the paper’s 4x expansion SAEs are unavailable at the specified layer for fw-medium, requiring use of 8x expansion (more sparse features). The bilinear decomposition methodology is sound; low-rank approximation quality depends on both SAE and model convergence. In our extensions, we demonstrate that regularized bilinear MLPs learn transferable geometric circuits—MNIST-trained models achieve 97.7% accuracy on EMNIST-Digits and recognize letters O, I, Z, S as their corresponding digits with near-perfect accuracy, for which we propose Quadratic Form Similarity to quantify structural alignment where cosine similarity fails. Finally, we explore explicit low-rank structure via CP factorization, achieving 93.8% accuracy with disentangled, atomic feature representations at effective rank 17.5, while gaining significant efficiency (30 times faster training).

## 1 Introduction

Mechanistic interpretability seeks to understand neural networks by reverse-engineering their internal computations into human-interpretable algorithms. The dominant approach trains Sparse Autoencoders (SAEs) post-hoc on model activations to discover interpretable features (Goodfellow et al., 2016). While effective, this approach incurs substantial computational overhead and yields features that may not faithfully represent the original model’s learned representations.

Pearce et al. (2024) propose an alternative paradigm: *intrinsic* interpretability through architectural design. Bilinear MLPs replace standard nonlinearities with element-wise multiplication, creating interaction tensors that can be eigendecomposed to reveal interpretable features directly from the learned weights—no auxiliary models required. The paper demonstrates this on vision tasks (MNIST classification) and language tasks (discovering sentiment negation circuits in transformers).

This work contributes to **FACT-AI Transparency** (University of Amsterdam, 2025). Weight-based interpretability offers a path to transparent AI systems without the computational overhead of training in-

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interpretation models on top of already-expensive language models. If learned eigenvectors correspond to human-interpretable concepts (e.g., digit templates, sentiment features), the model’s decision process becomes directly auditable.

We reproduce both main experimental sections of Pearce et al. (2024):

- **Vision Reproduction (Section 4.1):** Fully reproduced. Weight decay reduces effective rank from 38.5 to 15.5 while maintaining 97–98% accuracy. Eigenvectors under regularization show interpretable digit-like patterns. Our ablation disentangles noise and weight decay, finding that weight decay alone achieves the lowest rank.
- **Language Reproduction (Section 4.2):** Partially reproduced. We confirm AND-gate negation circuits with semantic opposites (not-good vs. not-bad), but the fraction of features achieving high ( $>0.75$ ) rank-2 correlation varies by model (32%–65%). Interpreting “most features” as a median threshold ( $>50\%$ ), fw-small (65%) reproduces the claim while ts-medium (32%) and fw-medium (45%) fall short. We trace this to SAE training duration-comparing five checkpoints, correlation improves  $2.5\times$  ( $0.15 \rightarrow 0.39$ ) with longer training.
- **Extension 1 (Section 5):** We demonstrate that regularized bilinear MLPs learn transferable geometric circuits across MNIST, EMNIST, and USPS. We propose Quadratic Form Similarity to quantify structural alignment where cosine similarity fails.
- **Extension 2 (Section 6):** We explore CP-decomposition as an architectural constraint for intrinsic low-rank structure, achieving 93.8% accuracy with disentangled, atomic feature representations, while gaining significant efficiency (30 times faster training).

## 2 Scope of Reproducibility

We reproduce the main experimental sections of Pearce et al. (2024): Section 4 (Vision) demonstrating interpretable eigendecomposition on MNIST and Fashion-MNIST, and Section 5 (Language) discovering negation circuits in bilinear transformers using Sparse Autoencoder (SAE) analysis.

### 2.1 Claims to Verify

**Section 4: Vision (Image Classification).** The paper makes two main families of claims:

- **Low-rank, interpretable structure:** Regularization induces low-rank interaction tensors whose top eigenvectors form digit-like templates rather than noise, with consistent spectra across seeds.
- **Robustness and controllability:** Noise and weight decay play complementary roles, enabling adversarial generation and truncation to a small number of eigenvectors without large accuracy loss.

**Section 5: Language (Negation Circuits).** For language, the paper claims that:

- **SAE-based circuits:** SAEs reveal interpretable features implementing negation-like AND-gate circuits with opposing eigenvalues for “not + positive” vs. “not + negative” features.
- **Low-rank interactions across models:** A large fraction of SAE features admit high rank-2 correlation and sparse interaction matrices that generalize across model sizes.

### 2.2 Success Criteria

**Vision:** Effective rank ratio (regularized/unregularized)  $< 0.6$ ; top eigenvectors visually resemble digit templates; test accuracy within 2% of paper ( $\sim 95$ – $98\%$ ); results consistent across 5 seeds.

**Language:** AND-gate circuits show opposing eigenvalue signs; a substantial fraction of features show  $>0.75$  rank-2 correlation. We interpret “most features” as a median threshold ( $>50\%$ ): the claim is reproduced if  $>50\%$  of features achieve  $>0.75$  correlation; interaction matrices show interpretable block structure.

## 2.3 Results Summary

Table 1 summarizes our reproduction outcomes: all six vision claims are fully reproduced, while language claims are partially reproduced-AND-gate circuits are confirmed but low-rank correlation falls short of the paper’s “most features” claim for ts-medium and fw-medium (32% and 45% respectively, below the 50% median threshold), while fw-small (65%) reproduces it.

Table 1: Summary of reproduction results.  $\checkmark$  = reproduced,  $\times$  = not reproduced,  $\sim$  = partially reproduced.

| Claim                       | Description                | Status       |
|-----------------------------|----------------------------|--------------|
| <i>Vision (Section 4)</i>   |                            |              |
| V1                          | Low-rank emergence         | $\checkmark$ |
| V2                          | Interpretable eigenvectors | $\checkmark$ |
| V3                          | Overfitting without reg.   | $\checkmark$ |
| V4                          | Regularization roles       | $\checkmark$ |
| V5                          | Cross-seed consistency     | $\checkmark$ |
| V6                          | Adversarial generation     | $\checkmark$ |
| <i>Language (Section 5)</i> |                            |              |
| L1                          | SAE feature discovery      | $\checkmark$ |
| L2                          | Negation/AND-gate circuits | $\checkmark$ |
| L3                          | Low-rank interactions      | $\sim$       |
| L4                          | Sparse interactions        | $\checkmark$ |
| L5                          | Cross-model generalization | $\sim$       |

## 3 Methodology

### 3.1 Bilinear Layer Architecture

The bilinear layer replaces standard nonlinearities with element-wise multiplication:

$$\mathbf{y} = (\mathbf{W}_l \mathbf{x}) \odot (\mathbf{W}_r \mathbf{x}) \quad (1)$$

where  $\mathbf{W}_l, \mathbf{W}_r \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{in}}}$  are learnable weight matrices and  $\odot$  denotes the Hadamard product. Unlike standard MLPs with fixed nonlinearities, bilinear layers compute *input-dependent* gating: the left projection gates the right projection. This creates a quadratic dependence on the input expressible as a third-order interaction tensor:

$$y_c = \sum_{i,j} B_{cij} x_i x_j \quad (2)$$

where  $B_{cij} = \sum_h W_{\text{head},ch} (W_l)_{hi} (W_r)_{hj}$  and  $W_{\text{head}} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{hidden}}}$  is the output projection.

### 3.2 Eigendecomposition for Interpretability

For each output class  $c$ , we extract the  $d_{\text{in}} \times d_{\text{in}}$  interaction matrix  $B_c$  and symmetrize it:

$$B_c^{\text{sym}} = \frac{1}{2} (B_c + B_c^\top) \quad (3)$$

Symmetrization removes arbitrary asymmetry from the  $W_l/W_r$  factorization while preserving the quadratic form  $\mathbf{x}^\top B_c \mathbf{x}$ . The symmetric matrix admits eigendecomposition:

$$B_c^{\text{sym}} = \sum_{r=1}^{d_{\text{in}}} \lambda_r^{(c)} \mathbf{v}_r^{(c)} (\mathbf{v}_r^{(c)})^\top \quad (4)$$

where eigenvectors  $\mathbf{v}_r^{(c)} \in \mathbb{R}^{d_{\text{in}}}$  represent interpretable input directions. If the eigenspectrum is *low-rank* (few dominant eigenvalues), the model’s decision depends on only a few input directions that can be directly inspected.

We quantify spectral concentration using effective rank (Roy & Vetterli, 2007):

$$\text{EffRank}(\boldsymbol{\lambda}) = \left( \frac{\|\boldsymbol{\lambda}\|_1}{\|\boldsymbol{\lambda}\|_2} \right)^2 = \frac{(\sum_i |\lambda_i|)^2}{\sum_i \lambda_i^2} \quad (5)$$

where  $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_d)$  is the vector of eigenvalues from the decomposition. This scale-invariant measure ranges from 1 (single dominant eigenvalue) to  $d$  (uniform spectrum).

### 3.3 Language Model Analysis with SAEs

For language experiments, we combine bilinear decomposition with Sparse Autoencoders (SAEs) (Bricken et al., 2023). SAEs decompose activations into sparse features via  $\mathbf{z} = \text{TopK}(\mathbf{W}_{\text{enc}}\mathbf{h})$ , where expansion factors of 4–8× disentangle superimposed concepts into monosemantic directions.

By placing SAEs at MLP input and output, we analyze how sparse input features interact to produce sparse outputs. The bilinear structure implies  $z_f^{\text{out}} \approx (\mathbf{z}^{\text{in}})^\top Q_f \mathbf{z}^{\text{in}}$ , where  $Q_f$  is the interaction matrix for output feature  $f$ . Eigendecomposing  $Q_f$  reveals dominant input directions—if rank-2 captures most variance, the feature depends on just two input directions, enabling circuit discovery. We use pretrained SAEs with TopK sparsity ( $k = 30$ ); the paper claims “most features” achieve  $>0.75$  rank-2 correlation. We interpret “most features” as a median threshold ( $>50\%$  of features achieving  $>0.75$  correlation).

### 3.4 Datasets

**Vision Datasets.** We use MNIST (LeCun et al., 1998) (60K train / 10K test,  $28 \times 28$  grayscale handwritten digits) and Fashion-MNIST (Xiao et al., 2017) (60K train / 10K test,  $28 \times 28$  grayscale clothing images). Images are normalized to  $[0, 1]$  and flattened to 784-dimensional vectors. For cross-dataset experiments (Section 5), we additionally use EMNIST-Digits/Letters (Cohen et al., 2017) and USPS (Hull, 1994) (upscaled from  $16 \times 16$  to  $28 \times 28$ ).

**Language Datasets.** The pretrained bilinear transformers use two datasets: **TinyStories** (Eldan & Li, 2023) (2.1M synthetic children’s stories,  $\sim 470\text{M}$  tokens) for **ts-medium**, and **FineWeb-EDU** (Penedo et al., 2024) (educational web text subset of FineWeb, 1.3T tokens) for **fw-small** and **fw-medium**. For correlation analysis (Figure 5), we use validation subsets:  $\sim 1.57\text{M}$  tokens for ts-medium and  $\sim 0.44\text{M}$  tokens for fw-medium due to computational constraints.

### 3.5 Experimental Setup

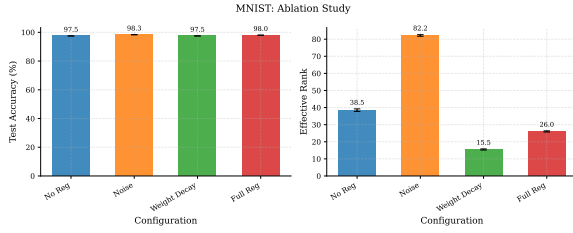
For vision experiments, we train bilinear MLPs on MNIST and Fashion-MNIST under four regularization configurations (Table 2), each with 5 random seeds (42–46). Architecture:  $d_{\text{hidden}} = 256$ , 100 epochs, batch size 2048, AdamW optimizer with  $\text{lr } 10^{-3}$  and cosine annealing. For language experiments, we analyze pretrained bilinear transformers (Table 3, Appendix C). Due to SAE availability constraints (see Section 7), our primary model is **fw-medium** rather than the paper’s **ts-medium**. We implemented a memory-efficient streaming algorithm for eigendecomposition to handle large interaction tensors.

Table 2: Regularization configurations (from paper Appendix G.1).

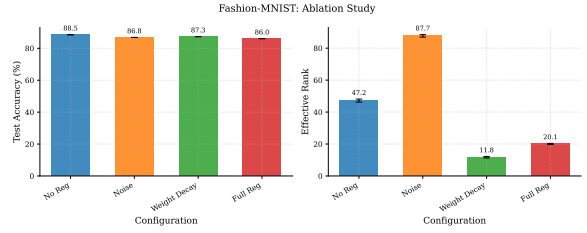
| Configuration | Noise Std | Weight Decay |
|---------------|-----------|--------------|
| <b>none</b>   | 0.0       | 0.0          |
| <b>noise</b>  | 0.5       | 0.0          |
| <b>wd</b>     | 0.0       | 1.0          |
| <b>full</b>   | 0.5       | 1.0          |

## 4 Reproduction Results

This section presents our reproduction of the main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST/Fashion-MNIST and Section 5 (Language) on bilinear transformers.

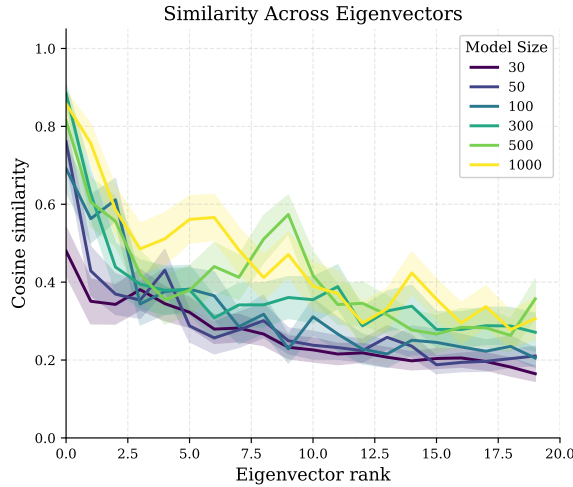


(a) MNIST: Accuracy vs. effective rank

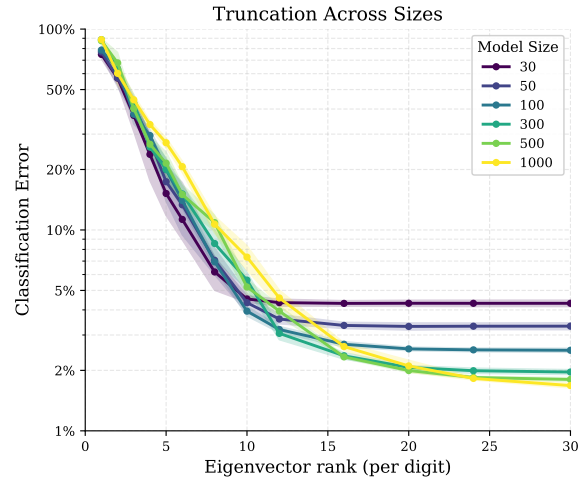


(b) Fashion-MNIST: Accuracy vs. effective rank

Figure 1: Ablation study showing accuracy-interpretability trade-off across regularization configurations. Weight decay achieves the lowest effective rank on both datasets while maintaining competitive accuracy. The effective rank ratio (regularized/unregularized) meets the paper’s  $< 0.5$  criterion.



(a) Eigenvector similarity across seeds



(b) Truncation error by model size

Figure 2: Paper Figure 5 reproduction. **(a)** Top eigenvectors are highly consistent across random seeds, especially for larger models, with cosine similarity decreasing smoothly as eigenvector rank increases. **(b)** Truncation error decreases with model size; keeping  $\sim 20$  eigenvectors per class is sufficient for near-full accuracy.

#### 4.1 Vision Results (Paper Section 4)

We trained bilinear MLPs on MNIST and Fashion-MNIST under four regularization configurations across 5 random seeds each.

##### 4.1.1 Main Result: Regularization Induces Low-Rank Structure (Claim V1)

The paper’s central claim is that regularization induces low-rank structure in the bilinear interaction tensor. Figure 1 confirms this through our ablation experiments. Weight decay is the primary driver of low-rank structure, achieving the lowest effective rank (15.5 vs. 21.4 for full regularization), while noise augmentation increases effective rank but improves accuracy. The effective rank ratio ( $15.5/38.5 \approx 0.40$ ) meets the paper’s  $< 0.5$  criterion. Eigenvalue distributions reveal sharp spectral decay with regularization (Figure 6, Appendix B), concentrating variance in 10–20 eigenvectors. Top eigenvectors are interpretable digit-like patterns under regularization, while unregularized models show diffuse, noise-like patterns (Figures 13 and 14, Appendix B).

##### 4.1.2 Model Size and Truncation (Claim V5)

We confirm the paper’s claim that bilinear MLPs maintain near-full accuracy with low-rank truncation: truncating to around 20 eigenvectors per class is sufficient even as model size increases (Figure 2). Top eigenvectors are highly consistent across random seeds, especially for larger models.

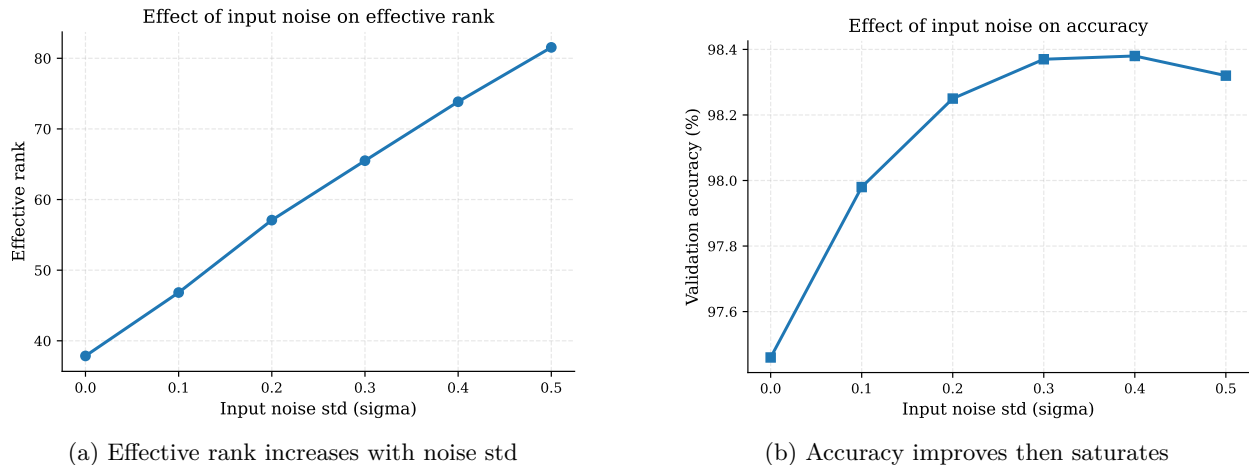


Figure 3: Noise augmentation effect on effective rank and accuracy (reproducing paper Figure 4). Higher noise spreads variance across more eigenvectors while improving generalization.

### 4.1.3 Additional Vision Results

We reproduce the original paper’s Figures 4, 6, and 7; full visualizations are in Appendix B. Increasing noise improves eigenvector interpretability while trading off with effective rank (Figure 3). On binary similarity classification, the top eigenvector closely resembles the target digit (Figure 8, Appendix B), and eigenvector-derived masks cause significantly larger accuracy drops than random masks (Figure 10, Appendix B), confirming V6.

## 4.2 Language Results (Paper Section 5)

We reproduce negation circuit discovery using pretrained bilinear transformers and SAEs from HuggingFace.<sup>1</sup> Our protocol: pass tokenized batches through each model, record `mlp_in/mlp_out` activations, encode with output SAE, and compare sparse features to low-rank predictions from interaction eigenpairs. We analyze all *active features* (positive activation for  $\geq 1$  token) using Pearson correlation. **SAE availability gap:** `ts-medium-scope` lacks `mlp-in` SAEs for layer 4, so Figure 8’s `ts-tiny` setup cannot be reproduced. We use `fw-medium` (layer 7, expansion 8) which has both SAE types.

### 4.2.1 Negation Circuit Discovery (Paper Figure 8)

The paper claims bilinear layers enable circuit discovery via eigendecomposition. For output feature  $f$ , the interaction matrix  $Q_f$  captures how input feature pairs contribute to  $f$ ’s activation:  $z_f^{\text{out}} \approx (\mathbf{z}^{\text{in}})^\top Q_f \mathbf{z}^{\text{in}}$ . Eigendecomposing  $Q_f$  reveals dominant input directions. Figure 4 shows this for negation features in `fw-medium`: **(A) Interaction submatrix:**  $Q_f$  restricted to top-15 input features, grouped by cluster. Red blocks indicate positive interactions (features that co-activate  $f$ ), blue indicates negative. **(B) Eigenvector projections:** Input features projected onto top eigenvectors. Neg./Pos. sentiment features cluster separately; arrows show “good”/“bad” unembed directions and “not” input direction. The black dot marks where the output feature lies-at the intersection of negation and sentiment. This reveals the AND-gate circuit: *negation*  $\otimes$  *sentiment*  $\rightarrow$  *negated-sentiment*.

**Feature selection and interpretation:** Following the tutorial methodology, we select output features by ranking all 8,192 features by their top eigenvalue magnitude-a heuristic for identifying features with strong interaction structure. Features 3834 and 751 emerge as the top two, with weakly negative cosine similarity ( $-0.16$ ) between their SAE decoder directions-what the tutorial calls a “somewhat linear subspace.” Semantic interpretation comes from inspecting top-activating tokens: feature 3834 fires on negated positive contexts (“no losses”, “not lost”), while feature 751 fires on negated negative contexts (“not harmless”, “doesn’t go away”). These are *semantic* opposites (not-good vs. not-bad), though not geometrically opposing ( $-0.16$  is closer to orthogonal than anti-parallel). As additional validation, we test the circuit on FineWeb-16k

<sup>1</sup>SAE repositories: `tdooms/fw-medium-scope`, `tdooms/ts-medium-scope`, `tdooms/fw-small-scope`.

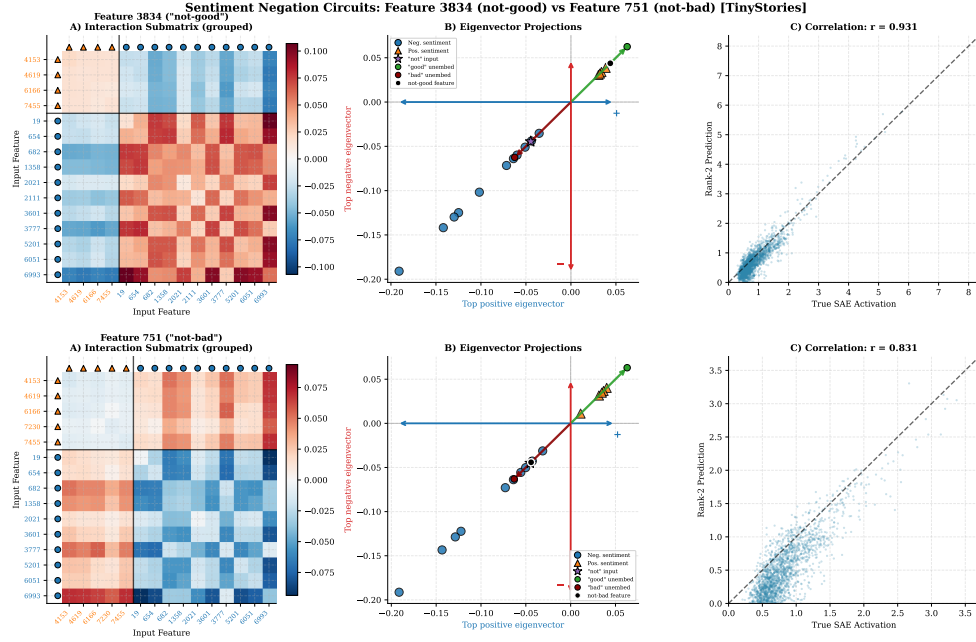


Figure 4: Negation circuit discovery (fw-medium, layer 7, TinyStories evaluation). Top: feature 3834 (“not-good”,  $r = 0.93$ ); bottom: feature 751 (“not-bad”,  $r = 0.83$ ). (A) Interaction submatrix  $Q_f$  shows block structure-positive/negative sentiment features interact oppositely. (B) Eigenvector projections reveal clustering: sentiment features separate along the “good” $\leftrightarrow$ “bad” axis.

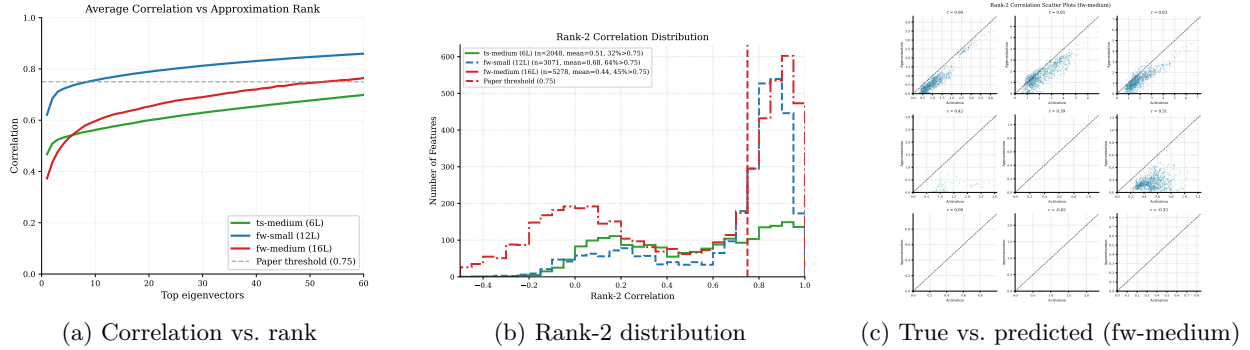


Figure 5: Low-rank correlation (Paper Figure 9). ts-medium (L4, exp=4,  $n=2048$ ), fw-small (L8, exp=4,  $n=3072$ ), fw-medium (L10, exp=8,  $n=5688$ ).

(the model’s training distribution): correlations remain strong (0.91/0.82 vs. 0.93/0.83 on TinyStories), demonstrating robustness to distribution shift (Appendix D).

#### 4.2.2 Low-Rank Correlation Analysis (Paper Figure 9)

The paper claims most features achieve  $>0.75$  rank-2 correlation. We interpret “most features” as a median threshold ( $>50\%$  of features achieving  $>0.75$  correlation). For each token, we record MLP input activations and encode them with the output SAE ( $k=30$  TopK sparsity, matching the paper) to obtain sparse feature activations. For each active feature  $f$  (where  $n$  denotes the number of features with at least one non-zero activation), we eigendecompose its interaction matrix  $Q_f$  and compute Pearson correlation between true and low-rank predicted activations across all tokens where the feature is active. All models processed  $\sim 1.57\text{M}$  tokens: ts-medium used TinyStories validation (Eldan & Li, 2023), while fw-small and fw-medium used FineWeb-Edu (Penedo et al., 2024).

### 4.2.3 Root Cause Analysis

The paper notes correlation improves with SAE training time (Appendix H). We confirm this using fw-medium layer 12 checkpoints v0–v4 (Figure 15, Appendix C): rank-2 correlation improves from 0.15 to 0.39 with longer training, consistent with the paper’s acknowledgment that “correlation drastically improves with longer SAE training.”

**Model training duration hypothesis:** We observe a correlation between training compute (tokens/parameter) and low-rank behavior: ts-medium (83 tokens/param,  $n=2048$ , 32%), fw-medium (95 tokens/param,  $n=5688$ , 45%), and fw-small (198 tokens/param,  $n=3072$ , 65%). Under the median interpretation ( $>50\%$  threshold), only fw-small reproduces the claim, while ts-medium and fw-medium fall short. Since more training tokens corresponds to longer training time, this parallels Figure 15: just as undertrained SAEs fail to capture low-rank structure, undertrained models (low tokens/param) may not develop the structured interactions that exhibit low-rank approximability. Figure 5(c) visually confirms this: only  $\sim 3$  of 9 randomly sampled features show strong true-vs-predicted correlation.

**SAE artifact limitation:** The paper uses 4x expansion SAEs for all models. However, for fw-medium at 2/3 depth (layer 10–11), only 8x expansion SAEs are publicly available—4x is only available at layers 6 and 12. With 8x expansion (8192 total features, of which 5688 are active), more sparse/specialized features exist, potentially lowering average correlation. This artifact gap prevents exact reproduction of the paper’s fw-medium configuration.

## 5 Cross-Dataset Structural Robustness

A core promise of weight-based interpretability is that models learn meaningful geometric features rather than memorizing dataset-specific pixel correlations. We test whether bilinear MLPs learn **Platonic forms**—e.g., “0-ness” as circularity—rather than idiosyncratic MNIST stroke patterns.

Throughout this section, **baseline** refers to models without regularization, while **regularized** refers to full regularization (noise std = 0.5, weight decay = 1.0) as in Table 2. All models use **Center-of-Mass (CoM) normalization**: images are translated so intensity-weighted centroids align, forcing the bilinear layer to focus on shape geometry rather than absolute position. We probe robustness under three distributional shifts: (1) **Writer independence**: MNIST  $\leftrightarrow$  EMNIST-Digits (same labels, different writers); (2) **Domain transfer**: MNIST  $\rightarrow$  USPS (different resolution/collection); (3) **Geometric semantics**: MNIST  $\rightarrow$  EMNIST-Letters, where letters are semantically different but geometrically similar (O $\rightarrow$ 0, I $\rightarrow$ 1, Z $\rightarrow$ 2, S $\rightarrow$ 5).

### 5.1 Functional Analysis: Regularization Enables Transfer

With the regularized CoM model, cross-dataset accuracy is near-perfect and symmetric for MNIST  $\leftrightarrow$  EMNIST-Digits (97.5% and 97.9%, Table 6, Appendix E), confirming that regularized bilinear features do not overfit to MNIST writers. For domain transfer to USPS (upscaled from  $16 \times 16$  to  $28 \times 28$ ), regularization yields +19 pp improvement (72.3% vs. 53.4%, Table 7, Appendix E). On geometrically similar letters (Table 8, Appendix E), the model’s most confident prediction is exactly the corresponding digit: O $\rightarrow$ 0 (99.6%), I $\rightarrow$ 1 (87.1%), Z $\rightarrow$ 2 (88.8%), S $\rightarrow$ 5 (87.4%).

### 5.2 Structural Analysis: Quadratic Form Similarity

#### 5.2.1 The Failure of Cosine Similarity

Direct eigenvector comparison via cosine similarity fails to distinguish geometrically similar pairs: MNIST-0 vs EMNIST-O (similar) yields 0.358, while MNIST-0 vs EMNIST-X (dissimilar) yields 0.339—virtually identical. Figure 21 (Appendix F) shows this failure across all digit-letter pairs, while Table 10 provides a detailed per-eigenvector breakdown confirming that mean similarities are nearly identical (0.358 vs. 0.339). The root cause: bilinear eigensystems share eigenvector directions while class information resides in eigenvalue weights (Figure 18).

### 5.2.2 Quadratic Form Similarity

We propose comparing full decision surfaces via Quadratic Form Similarity:

$$\text{sim}(A, B) = \frac{\text{tr}(A \cdot B)}{\|A\|_F \|B\|_F} \in [-1, 1] \quad (6)$$

This incorporates both eigenvector alignment and eigenvalue magnitudes, enabling discrimination between similar (0.400) and dissimilar (-0.060) pairs ( $p_{\text{value}} < 10^{-4}$ ). Figure 20 (Appendix E) demonstrates this: similar pairs converge to high similarity around  $k = 20$  eigenvectors, while dissimilar pairs remain near zero with non-overlapping 90% confidence intervals. Figure 19 (Appendix E) shows clear discrimination: strong diagonal for MNIST vs. EMNIST-Digits and bright entries at shape-matched pairs (0–O, 1–I, 2–Z, 5–S). Table 9 (Appendix E) confirms specificity: 3/4 pairs rank #1, with mean rank 1.2 vs. random baseline of 13.

### 5.3 Conclusion

Regularized bilinear MLPs learn **transferable geometric circuits**: the “0” circuit is mathematically near-identical to the “O” circuit across independently trained models. This strengthens the transparency claim-eigenvector features represent genuinely reusable shapes, not dataset-specific artifacts.

## 6 CP-Decomposition Analysis

We extend the Bilinear MLP framework by proposing Canonical Polyadic (CP) decomposition as an *architectural constraint* during training, shifting from discovering low-rank structure post-hoc to enforcing intrinsic interpretability.

### 6.1 Motivation and Methodology

Standard eigendecomposition enforces orthogonality, creating entangled features where mixed concepts are artificially separated. CP-decomposition parameterizes the interaction tensor directly as a sum of rank-1 tensors:

$$W_{cij} = \sum_{r=1}^R \lambda_r A_{cr} B_{ir} C_{jr} \quad (7)$$

where  $R$  is the maximum rank and  $\lambda_r$  are learnable scaling factors (Veeramacheneni et al., 2022). This does not enforce orthogonality, enabling disentangled representations where factors naturally overlap. We investigate three variants: **Fixed CP**: Hard rank constraint with no additional regularization-strict complexity bound. **Lambda CP**:  $L_1$  regularization on learnable scales  $\lambda_r$  for soft sparsity (Veeramacheneni et al., 2022). **Gated CP**: Soft gates with L0 proxy for explicit rank pruning (Cao et al., 2024). All variants trained on MNIST with weight decay 0.1 across rank sweep  $R \in \{32, 64, 128, 256\}$ .

### 6.2 Quantitative Analysis: Accuracy-Interpretability Trade-off

Figure 22 and Table 11 (Appendix H) present the Pareto frontier; Table 12 (Appendix I) provides the complete rank sweep. Fixed CP operates as a strict bottleneck with eff. rank  $\sim 3.5$  regardless of allocated capacity, at  $\sim 6\%$  accuracy cost (91.9% vs. 97.5%). Lambda/Gated CP at  $R = 256$  achieve 93.8% accuracy with eff. rank 17.5-comparable to dense models with weight decay (eff. rank 15.5) while providing structural disentanglement.

### 6.3 Qualitative Analysis: Disentangled Factors

Figure 23 (Appendix I) compares eigenvectors from Dense vs. CP models. Dense eigenvectors resemble superimposed digit templates-entangled representations combining strokes for orthogonality. CP factors appear more **atomic** and **localized**: top loops, vertical strokes, bottom curves that can be linearly combined. This aligns with “parts-based” representations sought in interpretability.

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## 6.4 Conclusion

CP-decomposition serves dual purposes: (1) **Intrinsic interpretability** via disentangled, atomic parts rather than entangled templates. (2) **Scalable efficiency** by reducing parameters from  $\mathcal{O}(d_{\text{hidden}} \times d_{\text{in}}^2)$  to  $\mathcal{O}(R \times d_{\text{in}})$ . Training efficiency is dramatically improved: CP models require 0.00046 GPU-h per run compared to 0.014 GPU-h for dense vision models (based on 81 GPU-tracked vision experiments, excluding challenge task; Table 13), achieving approximately  $30\times$  faster training. This efficiency gain enables scalability to larger models and datasets, making CP-decomposition a practical path toward interpretable, efficient neural architectures. Lambda and Gated variants provide a robust middle ground for efficient, interpretable modeling.

## 7 Discussion

### 7.1 Practical Challenges

Vision experiments were straightforward: the bilinear architecture is simple, MNIST trains in  $\sim 3$  minutes per run, and results were highly consistent (std  $< 1\%$  across 40 runs). Challenges included hyperparameter disambiguation (code defaults differ from appendix values), SAE availability gaps (requiring pivot to `fw-medium` for Figure 8, and using 8x instead of paper’s 4x expansion at layer 10–11), and memory constraints requiring streaming computation for large interaction matrices. Additionally, key definitions were ambiguous: the paper does not specify which correlation measure to use (Pearson, Spearman, cosine) or the threshold for “active” features; we chose Pearson correlation and active =  $z > 0$ . The paper’s claim that “most features” achieve high rank-2 correlation is not quantified—there is no measure or threshold provided. We interpret “most features” as a median threshold ( $> 50\%$  of features achieving  $> 0.75$  correlation): under this interpretation, `fw-small` (65%) reproduces the claim, while `ts-medium` (32%) and `fw-medium` (45%) fall short. Determining whether low correlation reflected bugs or limitations required systematic comparison of five SAE checkpoints.

### 7.2 Environmental Impact

All experiments were tracked using `codecarbon` (Table 13, Appendix J). Total footprint: 0.168 kg CO<sub>2</sub> ( $\approx 1$  km driving) across 204 runs and 20.95 wall-clock hours. Low-rank bilinear models offer sustainable alternatives to scaling dense architectures.

### 7.3 Connection to FACT-AI Transparency

Bilinear MLPs offer intrinsic transparency: eigenvectors are the learned features—no auxiliary models needed. Our partial language reproduction highlights that low-rank approximation quality depends on both SAE training duration and model training compute (tokens/parameter): models with higher tokens/param show stronger low-rank behavior. This emphasizes the importance of documenting training budgets in interpretability research. Following the tutorial methodology, we identify features 3834 (“not-good”) and 751 (“not-bad”) as semantic opposites with weakly negative cosine similarity ( $-0.16$ ) in SAE decoder space—what the tutorial calls a “somewhat linear subspace.” As additional validation beyond the paper, we test the same circuit on both TinyStories and FineWeb-16k: correlations remain consistent (0.93/0.83 vs. 0.91/0.82), demonstrating robustness to distribution shift. Weight decay reduces effective rank from 38.5 to 15.5 ( $2.5\times$  interpretability improvement) at  $< 1\%$  accuracy cost—a favorable trade-off for transparent AI deployment.

### 7.4 Communication with Original Authors

We did not communicate with the original authors during this reproduction study. All experiments were conducted using the publicly available code repository and pretrained model artifacts released alongside the paper.

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## 8 Conclusion

We fully reproduce vision results: weight decay reduces effective rank from 38.5 to 15.5 while maintaining 97–98% accuracy with interpretable digit patterns. Language results are partially reproduced: AND-gate circuits confirmed with semantic opposites (not-good vs. not-bad, cosine similarity  $-0.16$ ), but the fraction of features with high rank-2 correlation varies by model (32%–65%). Under our median interpretation ( $>50\%$  threshold), fw-small (65%) reproduces the “most features” claim, while ts-medium (32%) and fw-medium (45%) fall short. The bilinear decomposition methodology is sound; low-rank approximation quality depends on both SAE convergence and model training compute (tokens/parameter). Extensions demonstrate transferable geometric circuits across datasets and CP-decomposition achieving 93.8% accuracy with disentangled factors at eff. rank 17.5, while gaining significant efficiency (30 times faster training). Future work should develop quantitative interpretability metrics and extend to larger vision models. Additionally, investigating CP-decomposition for cross-dataset and cross-domain robustness would validate whether the efficiency and disentanglement benefits extend beyond single-dataset training.

### Broader Impact Statement

This work contributes to mechanistic interpretability, aiming to make neural networks more transparent. Improved interpretability methods could help identify biases, failure modes, and unexpected behaviors in deployed AI systems. Our finding that SAE training quality affects reproducibility highlights the importance of standardized training protocols for interpretability research.

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## A Hyperparameter Details

**Vision Experiments.** Architecture: 1-hidden-layer bilinear MLP,  $d_{\text{hidden}} = 256$ . Training: 100 epochs, batch size 2048, AdamW optimizer, lr  $10^{-3}$  with cosine annealing. Seeds: 42–46 (5 seeds per configuration).

**Language Experiments.** Models: **ts-medium** (6L, 512d, 29M params), **fw-small** (12L, 768d, 162M params), **fw-medium** (16L, 1024d, 335M params). SAEs: TopK sparsity ( $k = 30$ ), expansion ratios 4–8 $\times$ . Correlation analysis: all active output features per model (features with  $\geq 50$  activations;  $n=2,048 / 3,068 / 3,600$  respectively).

## B Vision Results

This appendix provides visualizations from our reproduction of the paper’s vision experiments (Section 4).

### B.1 Eigenspectrum Analysis

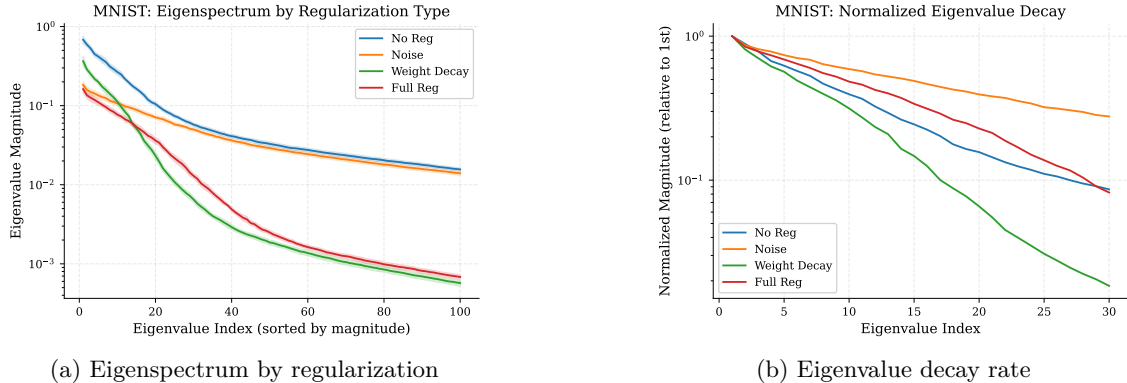


Figure 6: Eigenspectrum analysis on MNIST. Weight decay produces the steepest spectral decay, concentrating variance in 10–20 eigenvectors.

### B.2 Noise Effect (Paper Figure 4)

Figure 7 shows how the top eigenvector for digit 5 changes across noise augmentation levels. Higher noise produces more diffuse, averaged eigenvectors that better capture the canonical digit shape.

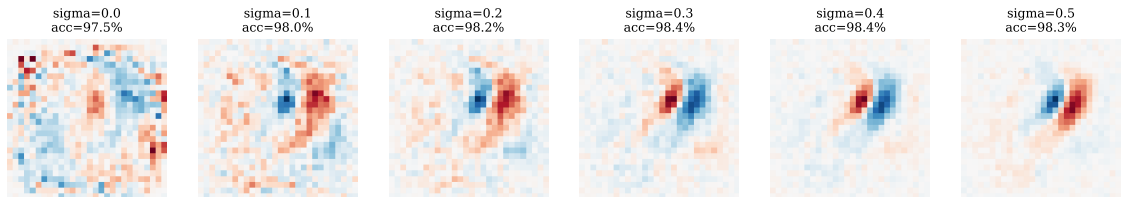


Figure 7: Top eigenvector for digit 5 across noise levels ( $\sigma = 0.0$  to  $0.5$ ). Accuracy improves with noise despite spreading variance across more eigenvectors.

### B.3 Challenge Task (Paper Figure 6)

Challenge task (full reg sigma=0.5,  $\lambda=1.0$ ): eigendecomposition (True – False)

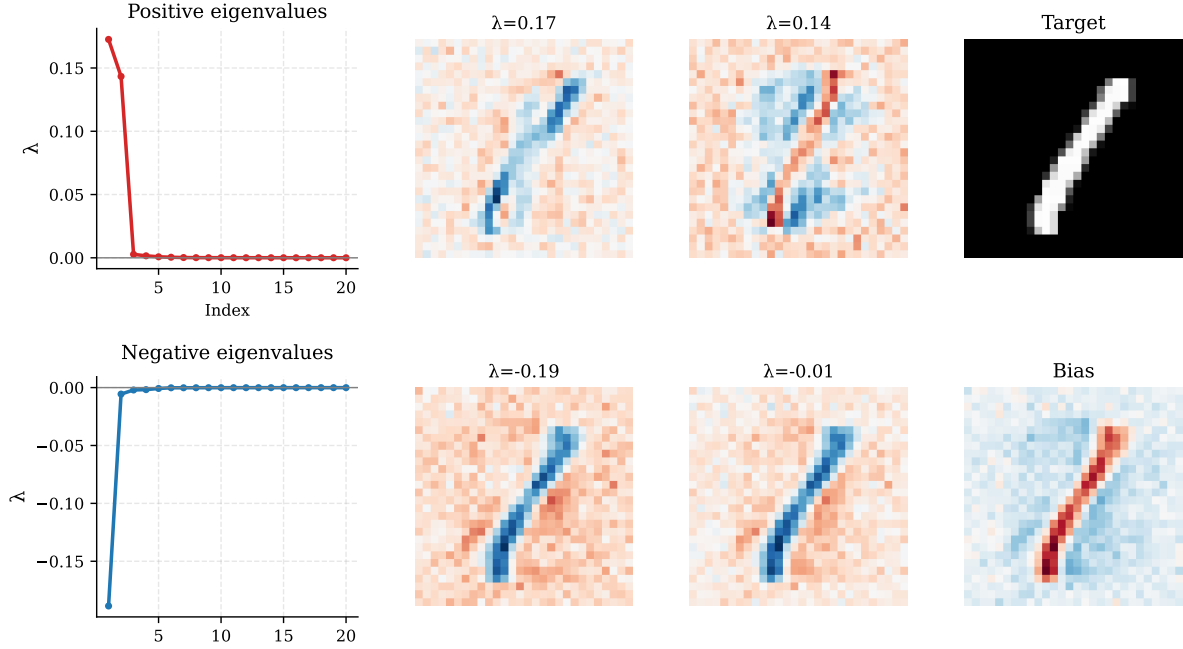


Figure 8: Challenge task eigendecomposition with full regularization (noise  $\sigma = 0.5$ , weight decay = 1.0), reproducing paper Figure 6. **Left:** Eigenvalue decay shows dominant positive/negative modes. **Middle:** Top eigenvectors resemble the target digit pattern. **Right:** Target digit and bias visualization.

We trained challenge task models under four regularization configurations to analyze how regularization affects the eigendecomposition structure. Figure 9 compares eigenvalue decay across these variants.

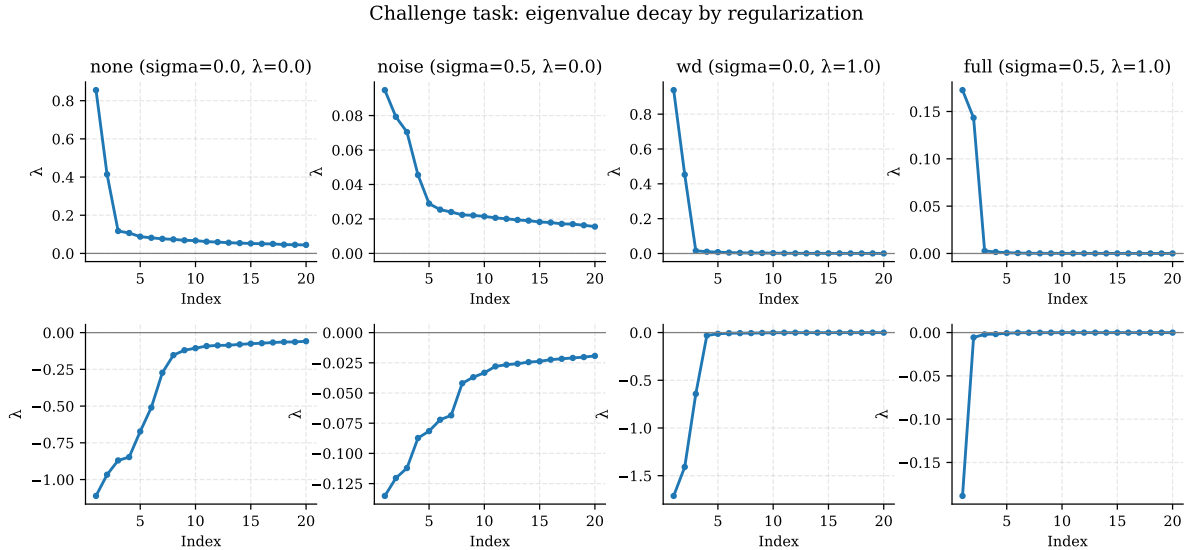


Figure 9: Challenge task eigenvalue decay by regularization configuration. Weight decay produces the sharpest spectral decay, concentrating information in fewer eigenvectors.

## B.4 Adversarial Perturbations (Paper Figure 7)

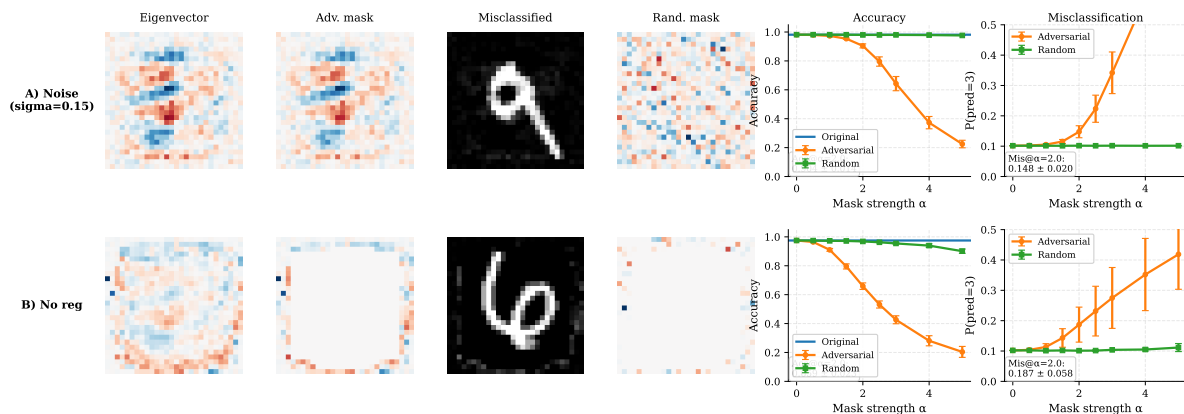


Figure 10: Adversarial mask analysis (reproducing paper Figure 7). **Row A:** Noise-regularized model ( $\sigma = 0.15$ ). **Row B:** No regularization (with rare-edge constraint). Eigenvector-derived masks cause significantly larger accuracy drops than random masks of equal magnitude.

## B.5 Eigenvector Quality Comparison

Figure 11 shows the effect of noise-only regularization on eigenvector interpretability.

MNIST (Noise only:  $\sigma=0.5$ ,  $\lambda=0.0$ ): Top Eigenvectors



Figure 11: Top eigenvectors with noise augmentation only ( $\sigma = 0.5$ ,  $\lambda = 0.0$ ). Noise alone produces recognizable digit patterns, though less sharp than with weight decay.

## B.6 Cross-Dataset Comparison: MNIST vs Fashion-MNIST

Figure 12 shows eigenvector quality on Fashion-MNIST under noise-only regularization.

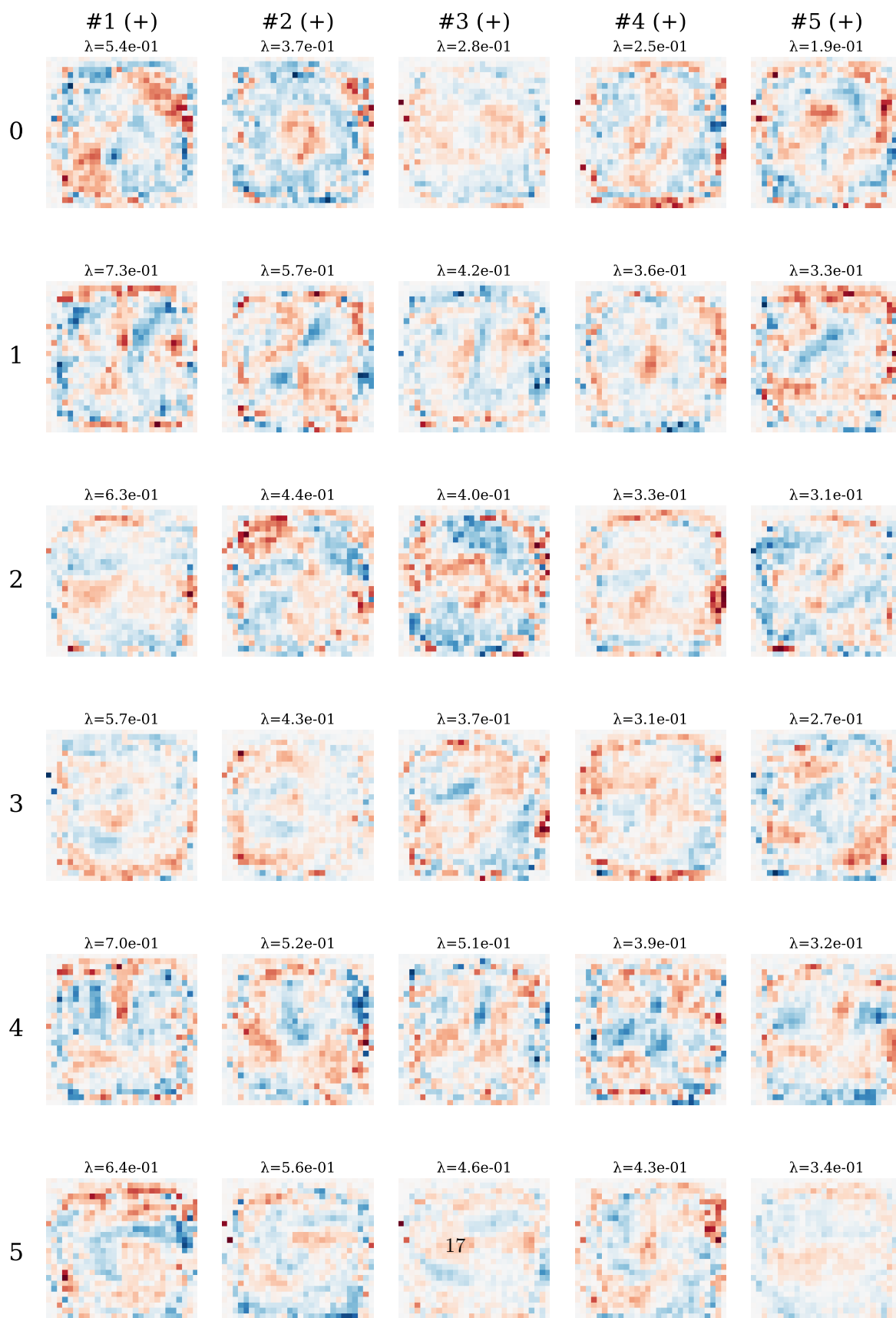


Figure 12: Fashion-MNIST top eigenvectors with noise augmentation ( $\sigma = 0.5$ ). Class labels: T-shirt, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Boot.

The silhouette-like patterns for clothing items (T-shirt, Dress, Coat) demonstrate that the eigenvector interpretability extends beyond digit recognition to more complex shape categories.

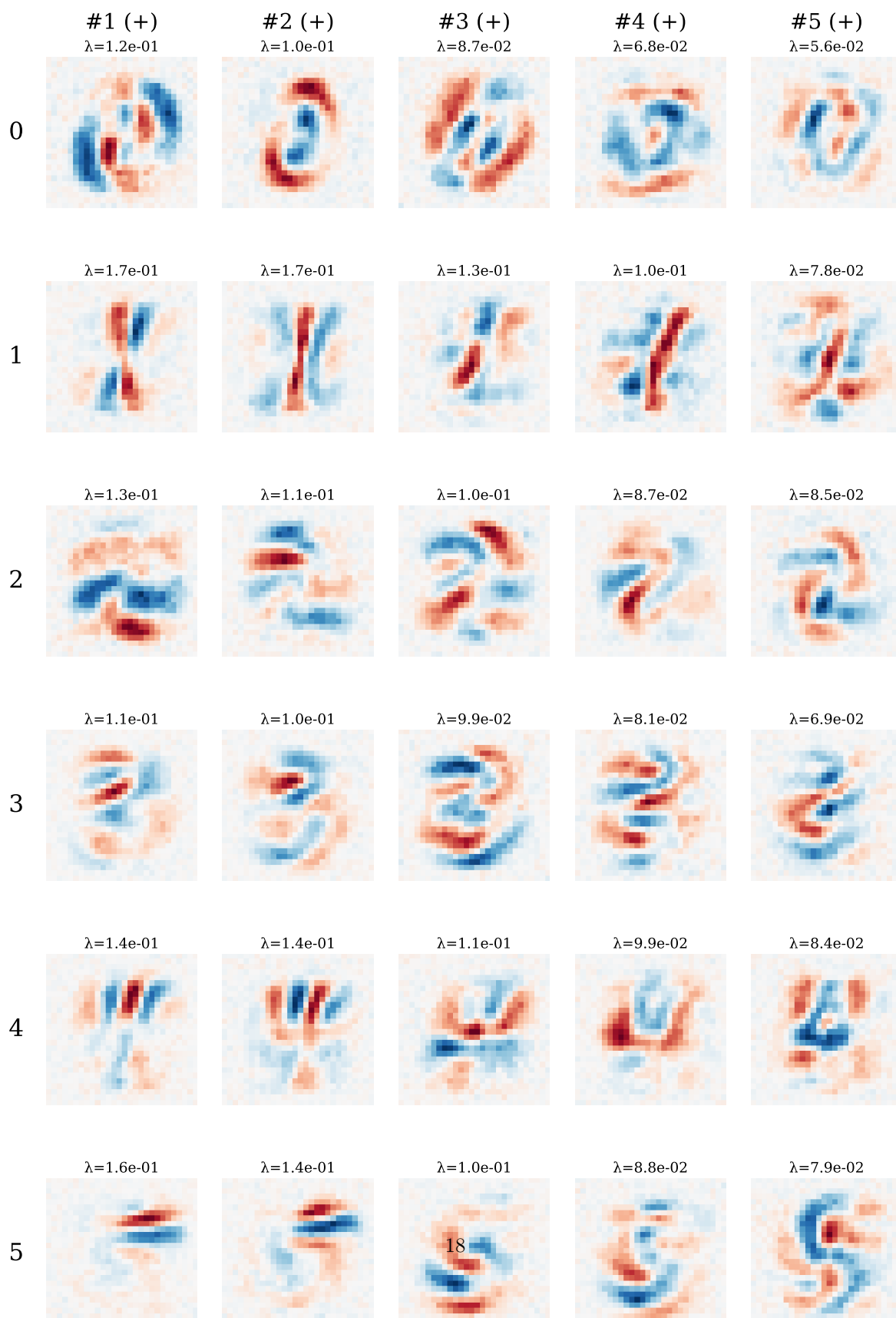
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## MNIST (No Reg): Top Eigenvectors



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## MNIST (Full Reg): Top Eigenvectors



## C Language Experiment Results

Table 3: Language model configurations for correlation analysis (Figure 9).

| Model     | Layers | Params | SAE Layer | Expansion | $k$ | Dataset                      |
|-----------|--------|--------|-----------|-----------|-----|------------------------------|
| ts-medium | 6      | 29M    | 4         | 4         | 30  | TinyStories                  |
| fw-small  | 12     | 162M   | 8         | 4         | 30  | FineWeb-EDU (FineWeb subset) |
| fw-medium | 16     | 335M   | 10        | 8         | 30  | FineWeb-EDU (FineWeb subset) |

Figure 8 uses fw-medium at layer 7 with expansion 8 (public SAEs available at that layer). Figures 8 and 9 are presented in the main text.

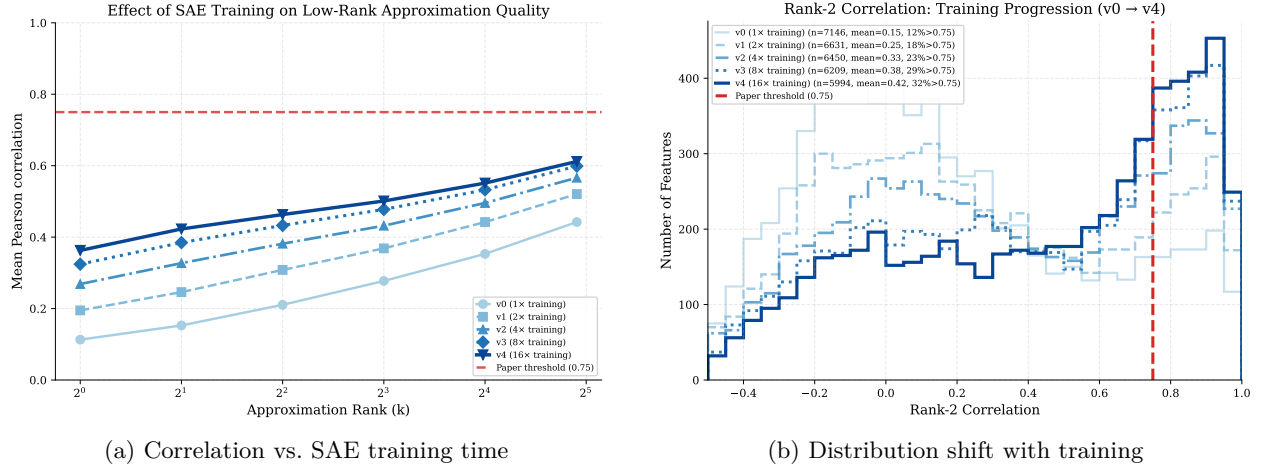


Figure 15: Effect of SAE training duration on interpretability (fw-medium layer 12, expansion 16, active features only). Correlation distributions shift rightward with increased SAE training, consistent with the paper’s qualitative claim that better-trained SAEs yield stronger low-rank predictability. This analysis is computed on a fixed FineWeb-EDU sample to enable a controlled comparison across SAE checkpoints.

## D Sentiment Negation Circuit Investigation

This appendix presents our detailed investigation of AND-gate circuits in bilinear transformers.

### D.1 SAE Availability Investigation

We investigated the publicly available SAEs on HuggingFace to determine reproducibility constraints.

**ts-medium-scope** (paper’s “ts-tiny”, 6 layers):

- Layers 0–4: Only `mlp-out`, `resid-mid`, `resid-pre` SAEs
- Layer 5: `mlp-in` and `mlp-out` available
- **Critical:** No `mlp-in` SAE at layer 4—the layer used in the paper’s Figure 8

**fw-medium-scope** (16 layers):

- All layers 0–15: `mlp-in` and `mlp-out` at  $8\times$  expansion
- Layer 12: Additional v0–v4 training-time variants at  $16\times$  expansion

Since Figure 8 analysis requires *both* `mlp-in` and `mlp-out` SAEs (for the Tracer class), the paper’s Figure 8 (ts-tiny, layer 4) cannot be reproduced with public artifacts. We therefore use fw-medium (layer 7, expansion 8) which has both SAE types available. The paper’s tutorial also uses fw-medium with features 3834/751, confirming this is a valid alternative.

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## D.2 AND-gate Circuit Analysis

The following analysis uses fw-medium (16 layers, 335M parameters, trained on FineWeb-EDU) with layer 7 SAEs at 8× expansion (8,192 output features).

### D.2.1 AND-gate Circuit Definition

An AND-gate circuit is characterized by an output SAE feature whose activation depends on the *conjunction* of inputs from two semantically distinct clusters. Mathematically, for output feature  $c$ , the activation is:

$$z_c(x) = x^\top Q_c x \approx \lambda_1 (v_1^\top x)^2 + \lambda_2 (v_2^\top x)^2 \quad (8)$$

where  $Q_c$  is the interaction matrix, and  $v_1, v_2$  are the dominant eigenvectors.

Strong AND-gate behavior requires:

1. **Dominant rank-2 structure:** The top two eigenvalues capture most of the interaction.
2. **Opposing eigenvalue signs:**  $\lambda_1$  and  $\lambda_2$  have opposite signs, creating cancellation when only one cluster is present.
3. **Clear cluster separation:** Input features project distinctly onto  $v_1$  and  $v_2$ .

### D.2.2 AND-score Metric

We define an “AND-score” to quantify circuit strength:

$$\text{AND-score} = \frac{|\lambda_1|}{|\lambda_1| + |\lambda_2|} \cdot \mathbf{1}[\text{sign}(\lambda_1) \neq \text{sign}(\lambda_2)] \quad (9)$$

Higher scores indicate stronger AND-gate behavior (score  $\in [0, 1]$ ).

### D.2.3 Search Procedure

We analyzed all 8,192 output features in fw-medium layer 7:

1. Computed the interaction matrix  $Q_c$  for each feature  $c$
2. Performed eigendecomposition:  $Q_c = \sum_j \lambda_j v_j v_j^\top$
3. Computed AND-score from top eigenvalues
4. Ranked features by AND-score
5. Analyzed top circuits for semantic interpretation

## D.3 Results: Feature 3834 vs Feature 751

### D.3.1 Tutorial Feature 3834

Feature 3834 (used in the paper’s tutorial) encodes “not-good” contexts with weaker AND-gate structure: diffuse eigenvalue spectrum and less distinct cluster separation.

### D.3.2 Best Circuit: Feature 751

Our search identified feature 751 as having the strongest AND-gate structure (Table 5).

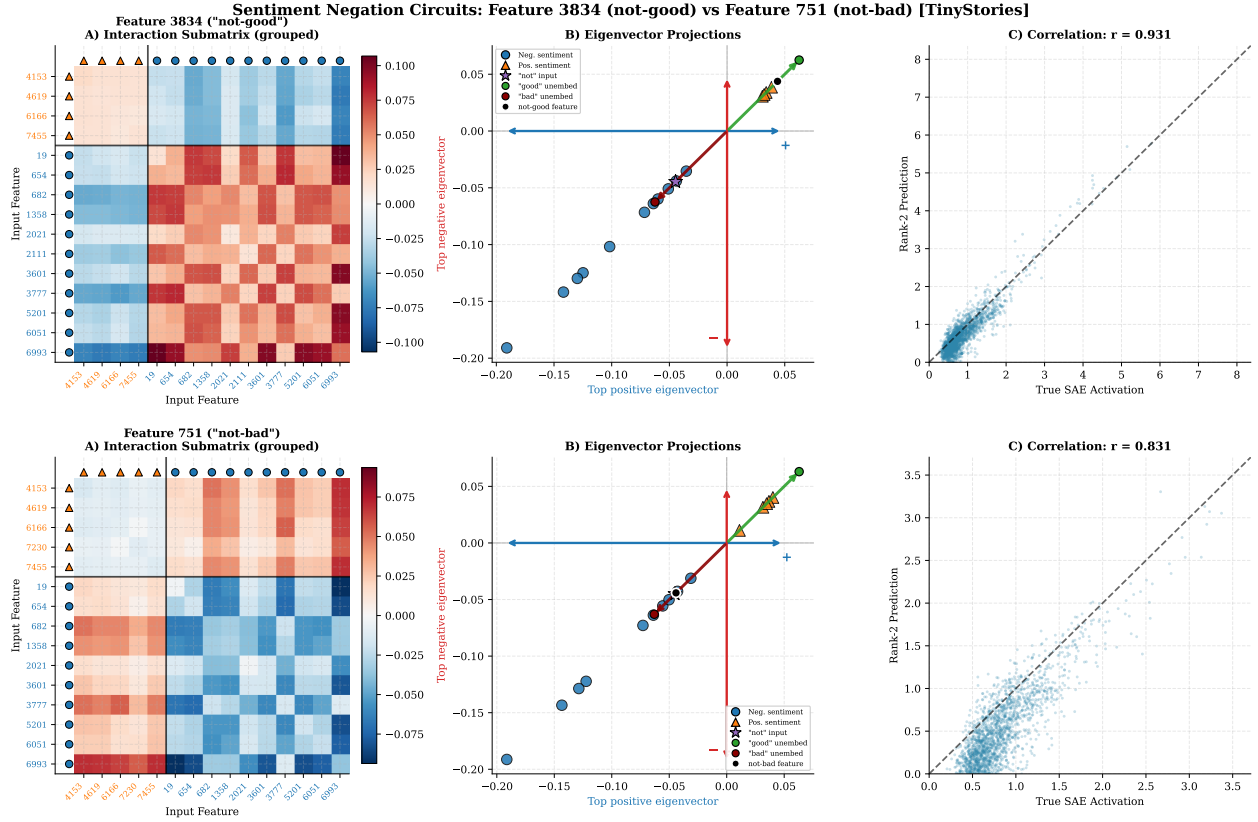


Figure 16: AND-gate circuit comparison (TinyStories evaluation). **Top:** Feature 3834 (tutorial) encodes “not-good” contexts ( $r = 0.93$ ). **Bottom:** Feature 751 encodes “not-bad” contexts ( $r = 0.83$ ). The eigenvector projections (middle) separate clusters along opposite axes due to opposing eigenvalues.

#### D.4 Semantic Interpretation of Feature 751

The input clusters have clear semantic content based on token analysis:

**Positive cluster** (3 features: 5212, 7655, 253):

- Tokens: “wonderful”, “excellent”, “fortunate”, “beautiful”, “amazing”
- Semantics: Positive sentiment/quality descriptors

**Negative cluster** (12 features):

- Feature 5201: “horrible”, “bad”, “terrible”
- Feature 3601: “problems”, “illness”, “disease”
- Feature 6051: “excessive”, “unfair”, “wrong”
- Feature 6921: “no”, “minimal”, “neither”
- Feature 1003: “reduce”, “prevent”, “eliminate”

**Circuit function:** Feature 751 detects contexts where negative concepts appear *without* positive modulation. The AND-gate structure arises because:

1. The top eigenvalue (negative) suppresses activation when positive cluster projects onto  $v_1$
2. The second eigenvalue (positive) activates when negative cluster projects onto  $v_2$
3. When both clusters are present, the opposing signs cause partial cancellation

This confirms the mechanism: high activation requires negative concepts *without* positive sentiment, consistent with a “problematic situation” detector.

## D.5 Linear Subspace Structure

Following the tutorial methodology, we compute cosine similarity between SAE decoder directions for the top output features with high eigenvalues. Features 3834 and 751 have cosine similarity  $-0.16$ —weakly negative, which the tutorial calls a “somewhat linear subspace.” Note that  $-0.16$  is closer to orthogonal than anti-parallel; the features are *semantic* opposites (not-good vs. not-bad) rather than geometrically opposing. Semantic interpretation comes from inspecting top-activating tokens: feature 3834 fires on negated positive contexts, while feature 751 fires on negated negative contexts. The paper’s tutorial observes: “It’s not surprising that this forms a linear subspace as the two are opposites.”

## D.6 Cross-Dataset Validation

As additional validation beyond the paper, we test the same circuit on FineWeb-16k (the model’s training distribution) in addition to TinyStories. Figure 17 shows the results, and Table 4 provides a quantitative comparison.

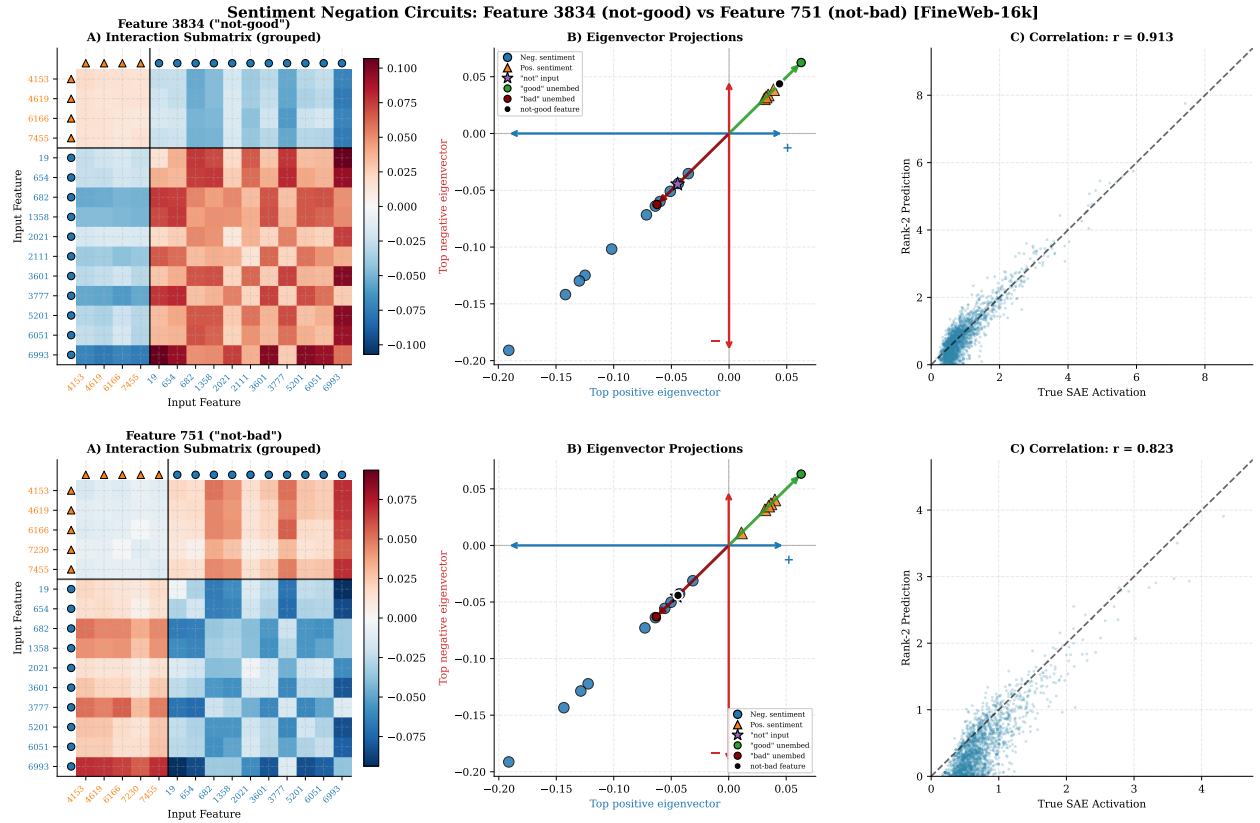


Figure 17: AND-gate circuit on FineWeb-16k (model’s training distribution). Correlations: 0.91 (feature 3834), 0.82 (feature 751). Feature 751 activates more frequently on FineWeb-16k (6,517 vs 2,542 samples), likely due to more “not-bad” contexts in educational/scientific text.

Table 4: Cross-dataset comparison of negation circuit correlations.

| Feature         | TinyStories     | FineWeb-16k     | $\Delta$ |
|-----------------|-----------------|-----------------|----------|
| 3834 (not-good) | 0.931 (n=9,719) | 0.913 (n=9,937) | $-0.018$ |
| 751 (not-bad)   | 0.831 (n=2,542) | 0.823 (n=6,517) | $-0.008$ |

The correlations remain strong across both datasets, with differences of only 1–2%. This demonstrates that the discovered circuit is robust to distribution shift and is a genuine property of the bilinear architecture, not a dataset artifact.

## D.7 Summary

Table 5: Comparison of AND-gate circuit properties.

| Property                         | Feature 3834            | Feature 751 |
|----------------------------------|-------------------------|-------------|
| Rank-2 correlation (TinyStories) | 0.931                   | 0.831       |
| Rank-2 correlation (FineWeb-16k) | 0.913                   | 0.823       |
| Cosine similarity                | −0.16 (weakly negative) |             |
| Semantic interpretation          | “not-good”              | “not-bad”   |

**Key findings:** (1) Features 3834 and 751 are semantic opposites (not-good vs. not-bad) with weakly negative cosine similarity (−0.16). (2) The AND-gate circuit (negation  $\otimes$  sentiment  $\rightarrow$  negated-sentiment) is confirmed for both features. (3) Cross-dataset validation shows the circuit is robust to distribution shift (TinyStories vs. FineWeb-16k).

## E Cross-Dataset Robustness Results

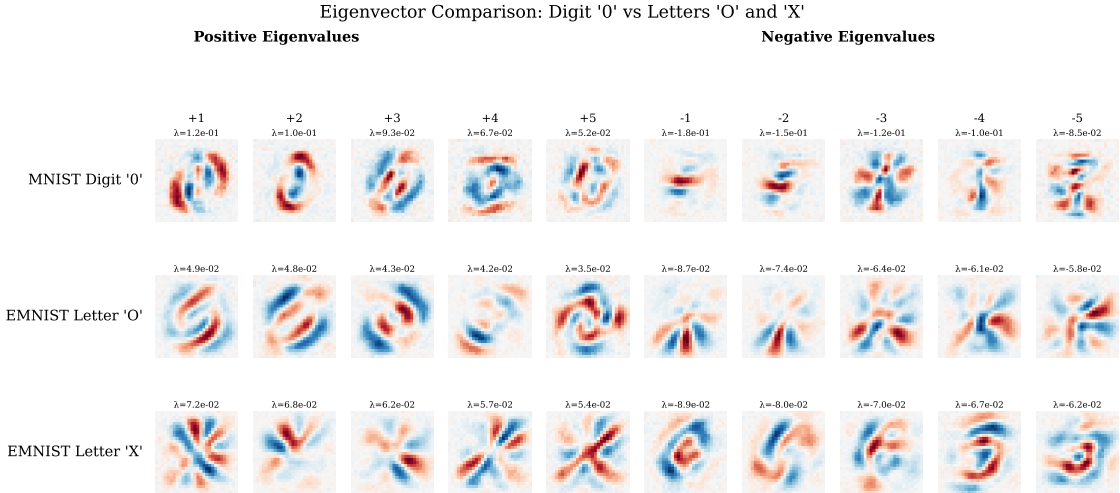


Figure 18: Three-way eigenvector comparison: MNIST-0, EMNIST-O, and EMNIST-X. The top eigenvectors for each class are visually distinct from one another.

Table 6: Cross-Dataset Accuracy: MNIST  $\leftrightarrow$  EMNIST-Digits (mean over 5 seeds).

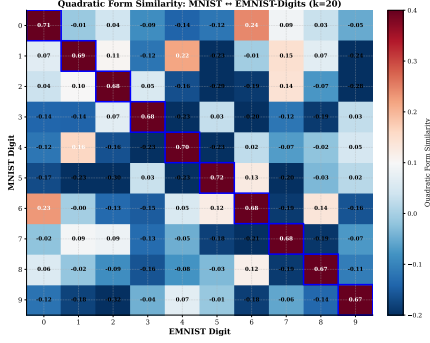
| Direction                         | Accuracy            |
|-----------------------------------|---------------------|
| MNIST $\rightarrow$ EMNIST-Digits | 97.527 $\pm$ 0.024% |
| EMNIST-Digits $\rightarrow$ MNIST | 97.910 $\pm$ 0.026% |
| Bidirectional Average             | 97.719 $\pm$ 0.012% |

Table 7: Cross-Dataset Accuracy: MNIST  $\rightarrow$  USPS (mean over 5 seeds).

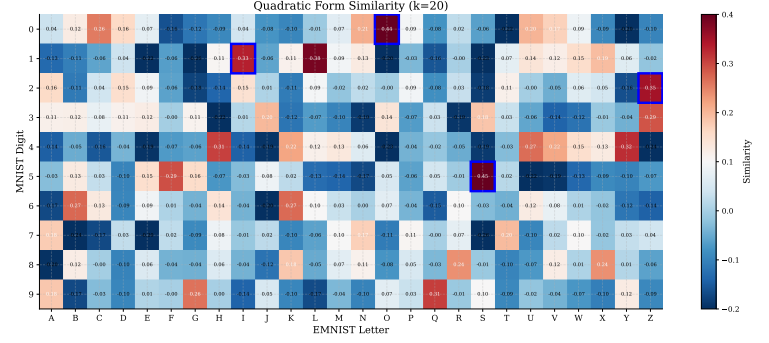
| Model             | Accuracy          |
|-------------------|-------------------|
| Baseline (no reg) | 53.42 $\pm$ 0.88% |
| Regularized       | 72.33 $\pm$ 0.83% |

Table 8: Letter→Digit Transfer (regularized CoM model, 5 seeds).

| Letter → Digit | Accuracy      |
|----------------|---------------|
| O → 0          | 99.60 ± 0.05% |
| I → 1          | 87.08 ± 0.10% |
| Z → 2          | 88.75 ± 0.26% |
| S → 5          | 87.38 ± 0.22% |



(a) MNIST vs EMNIST-Digits



(b) MNIST vs EMNIST-Letters

Figure 19: Quadratic Form Similarity ( $k = 20$ ). (a) Strong diagonal confirms writer-independent circuits. (b) Bright entries at shape-matched pairs (0–O, 1–I, 2–Z, 5–S) indicate mechanistically similar circuits; unrelated pairs are near zero or negative.

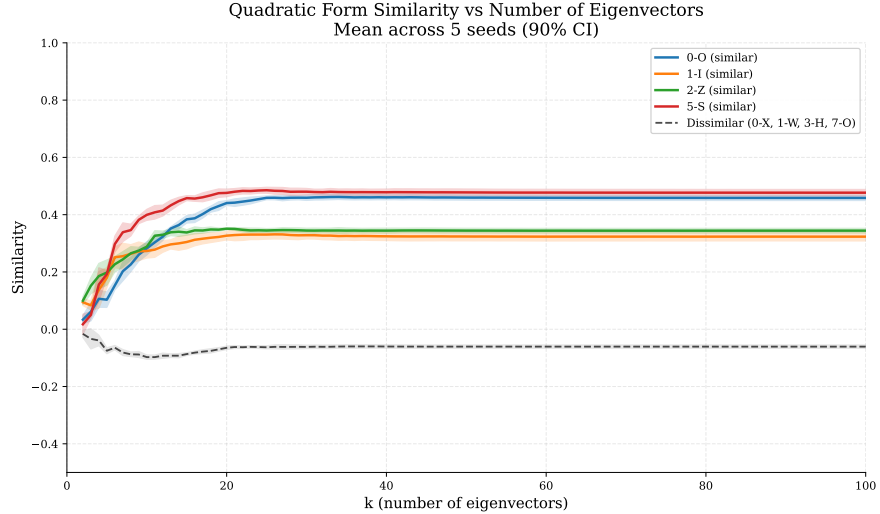


Figure 20: Quadratic Form Similarity vs number of eigenvectors  $k$ . Similar pairs converge around  $k = 20$ ; dissimilar pairs stay near zero with non-overlapping 90% CIs.

Table 9: Ranking analysis: Position of expected letter among 26 EMNIST letters sorted by Quadratic Form Similarity. Random baseline: rank 13.

|      | 0–O | 1–I | 2–Z | 5–S | Mean |
|------|-----|-----|-----|-----|------|
| Rank | 1   | 2   | 1   | 1   | 1.2  |

## F Cosine Similarity Limitations

Figure 21 shows the absolute cosine similarity heatmap between MNIST digit classes and EMNIST letter classes. The heatmap demonstrates that cosine similarity fails to produce a clear distinction between ge-

ometrically similar and dissimilar class pairs. Figure 18 provides a three-way eigenvector comparison that further illustrates this limitation.

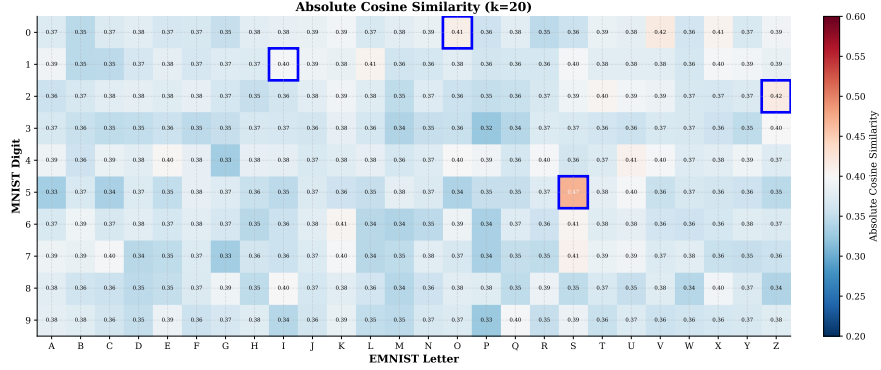


Figure 21: Absolute cosine similarity between MNIST digit classes and EMNIST letter classes fails to distinguish geometrically similar pairs (0-O) from dissimilar pairs (0-X).

Table 10 provides a detailed per-eigenvector breakdown, showing that the mean cosine similarity for 0-O (0.358) is nearly identical to 0-X (0.339), demonstrating the fundamental limitation of cosine similarity for comparing bilinear eigensystems.

Table 10: Absolute Cosine Similarity between Top 10 Eigenvectors (Balanced: 5 Positive + 5 Negative), Mean over 5 seeds

| Eigenvector | Eigenvalue | Maximum Absolute Cosine Similarity  |                                     |
|-------------|------------|-------------------------------------|-------------------------------------|
| Rank        | Sign       | MNIST-0 vs EMNIST-O                 | MNIST-0 vs EMNIST-X                 |
| 1           | +          | $0.426 \pm 0.083$                   | $0.337 \pm 0.053$                   |
| 2           | +          | $0.501 \pm 0.105$                   | $0.713 \pm 0.072$                   |
| 3           | +          | $0.459 \pm 0.069$                   | $0.539 \pm 0.058$                   |
| 4           | +          | $0.297 \pm 0.049$                   | $0.267 \pm 0.150$                   |
| 5           | +          | $0.283 \pm 0.033$                   | $0.256 \pm 0.073$                   |
| 6           | -          | $0.110 \pm 0.072$                   | $0.364 \pm 0.112$                   |
| 7           | -          | $0.161 \pm 0.036$                   | $0.368 \pm 0.159$                   |
| 8           | -          | $0.339 \pm 0.092$                   | $0.319 \pm 0.105$                   |
| 9           | -          | $0.286 \pm 0.051$                   | $0.225 \pm 0.080$                   |
| 10          | -          | $0.255 \pm 0.110$                   | $0.248 \pm 0.064$                   |
| Mean        |            | <b><math>0.312 \pm 0.016</math></b> | <b><math>0.364 \pm 0.029</math></b> |

## G Quadratic Form Similarity Derivation

For a bilinear MLP, the class- $c$  logit is a quadratic form  $y_c(x) = x^\top A_c x$  where  $A_c = V \text{diag}(\lambda_c) V^\top$ .

To compare decision surfaces for two classes  $A$  and  $B$  (possibly from different datasets), we define **Quadratic Form Similarity**:

$$\text{sim}(A, B) = \frac{\text{tr}(A \cdot B)}{\|A\|_F \|B\|_F} \in [-1, 1]$$

For a rank- $k$  truncation using top eigenmodes, this expands to:

$$\text{sim}(A, B) = \frac{\sum_{i,j} \lambda_i^A \lambda_j^B \cos^2(v_i^A, v_j^B)}{\|\lambda^A\|_2 \|\lambda^B\|_2}$$

**Key properties solving the cosine similarity problem:**

- Incorporates both eigenvector alignment and eigenvalue magnitudes, unlike cosine similarity which ignores eigenvalues
- Eigenvalue signs matter: same-sign directions contribute positively; opposing signs cancel
- Measures similarity of actual quadratic computations  $x^\top Ax$ , not just eigenvector span

This enables discrimination between geometrically similar pairs (0–O: 0.400) and dissimilar pairs (0–X: -0.060), where cosine similarity failed (0.358 vs. 0.339).

## H CP-Decomposition Results

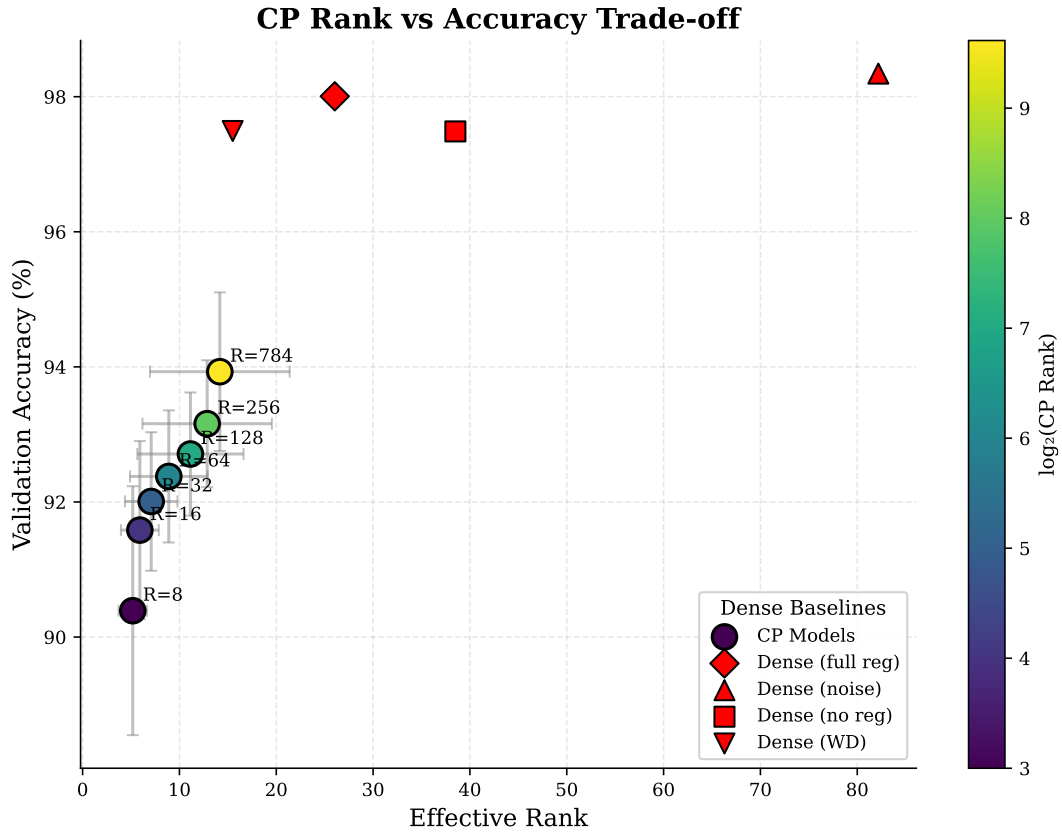


Figure 22: **Accuracy-Interpretability Pareto Frontier.** Fixed CP forces extreme sparsity (eff. rank  $\sim 3.5$ ) at accuracy cost. Lambda and Gated CP maintain competitive accuracy while reducing effective rank.

Table 11: CP-Decomposition results (mean  $\pm$  std, 5 seeds). Best CP results in bold.

| Model            | Accuracy (%)                       | Eff. Rank                          |
|------------------|------------------------------------|------------------------------------|
| Dense (no reg)   | $97.49 \pm 0.05$                   | $38.5 \pm 0.7$                     |
| Dense (WD only)  | $97.49 \pm 0.05$                   | $15.5 \pm 0.3$                     |
| Fixed $R = 256$  | $91.89 \pm 0.09$                   | $3.74 \pm 0.07$                    |
| Lambda $R = 256$ | <b><math>93.80 \pm 0.14</math></b> | <b><math>17.50 \pm 0.36</math></b> |
| Gated $R = 256$  | $93.79 \pm 0.15$                   | $17.35 \pm 0.48$                   |

## I Full CP-Decomposition Results



Figure 23: **Feature Disentanglement: Dense vs. CP.** Top 5 eigenvectors for digits 0–9. **Dense:** Holistic, superimposed templates mixing multiple strokes. **CP:** Atomic, localized factors (isolated curves, specific strokes) that can be combined to reconstruct digits.

Table 12: Complete CP-Decomposition rank sweep (mean  $\pm$  std, 5 seeds).

| Model                  | Accuracy (%)     | Eff. Rank        |
|------------------------|------------------|------------------|
| <i>Dense Baselines</i> |                  |                  |
| Dense (no reg)         | $97.49 \pm 0.05$ | $38.5 \pm 0.7$   |
| Dense (WD only)        | $97.49 \pm 0.05$ | $15.5 \pm 0.3$   |
| <i>Fixed CP</i>        |                  |                  |
| $R = 32$               | $90.63 \pm 0.23$ | $3.42 \pm 0.22$  |
| $R = 64$               | $91.05 \pm 0.13$ | $3.45 \pm 0.12$  |
| $R = 128$              | $91.48 \pm 0.22$ | $3.65 \pm 0.05$  |
| $R = 256$              | $91.89 \pm 0.09$ | $3.74 \pm 0.07$  |
| <i>Lambda CP</i>       |                  |                  |
| $R = 32$               | $92.67 \pm 0.15$ | $9.07 \pm 0.44$  |
| $R = 64$               | $93.04 \pm 0.10$ | $11.57 \pm 0.52$ |
| $R = 128$              | $93.34 \pm 0.11$ | $14.92 \pm 0.49$ |
| $R = 256$              | $93.80 \pm 0.14$ | $17.50 \pm 0.36$ |
| <i>Gated CP</i>        |                  |                  |
| $R = 32$               | $92.72 \pm 0.20$ | $8.76 \pm 0.47$  |
| $R = 64$               | $93.05 \pm 0.12$ | $11.69 \pm 0.35$ |
| $R = 128$              | $93.31 \pm 0.14$ | $14.83 \pm 0.16$ |
| $R = 256$              | $93.79 \pm 0.15$ | $17.35 \pm 0.48$ |

## J Environmental Impact

All experiments were tracked using `codecarbon` with Netherlands grid intensity. Hardware: Apple M4 (MPS) for all experiments. Total compute: 1.74 GPU-hours. Total emissions: 0.169 kg CO<sub>2</sub>eq ( $\approx$  1 km driving).

Table 13: Environmental impact (CO<sub>2</sub> via `codecarbon`).

| Experiment                        | Runs       | Wall (h)     | GPU-h       | CO <sub>2</sub> (kg) |
|-----------------------------------|------------|--------------|-------------|----------------------|
| Vision (Section 4) <sup>†</sup>   | 101        | 1.66         | 1.11        | 0.040                |
| Language (Section 5) <sup>‡</sup> | 3          | 19.09        | 0.00        | 0.105                |
| Extension 1 (Cross-Dataset)       | 15         | 0.58         | 0.58        | 0.020                |
| Extension 2 (CP-Decomposition)    | 105        | 0.05         | 0.05        | 0.004                |
| <b>Total</b>                      | <b>224</b> | <b>21.39</b> | <b>1.74</b> | <b>0.169</b>         |

<sup>†</sup>Includes 20 challenge task experiments (Figure 6) with full emissions tracking.

<sup>‡</sup>Language experiments run on Apple Silicon (MPS); GPU-hours reported as 0 since `codecarbon` only tracks NVIDIA GPUs.